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CHAPTER 1

INTRODUCTION

The unit hydrograph concept based hydrological model is most useful for estimating the runoff hydrograph of a catchment area. To draw unit hydrograph of a catchment complete data regarding precipitation and flood hydrograph are required. In the case of developing countries like India, the gauging stations are rare and even the rainfall data is very scanty. In major of the basins particularly those which are distant locations, facts and figures regarding the precipitation would be low-cut. In order to develop unit hydrograph for such catchments, empirical relations between the basin characters and the flow characters are developed. Unit hydrographs developed from such relations are called synthetic unit hydrograph. The examples of some of these methods are Snyder's Method, Taylor and Schwarz Model, Soil Conservation Service Method and others. These methods utilize some empirical relations equating between the physiographic characters of the basin and the parameters of the unit hydrograph such as the basin lag, base width, peak discharge and UH widths. The ease with the development of unit hydrograph from these methods with fewer data and giving smooth results makes them employable. Generally, river flow and flood are not available in many areas for planning and designing hydraulic structures and other management facilities for water. These situations are common in India due to lack of gauging stations in many locations. Thus, development of synthetic unit hydrograph is necessary for developing flood hydrograph, in watershed simulation models and in studies of forecasting and flood warning systems.

1.1 Gauged and Ungauged Catchments

Rainfall is essentially measured based on the vertical height of water that would collect on a level surface if the rainfall is retained where it fell. The rainfall is generally measured in millimetres per hour (mm/h). a relatively small area is taken for the purpose of the measurement of the rainfall and the volume of rain water collected over that area divide by the area gives the depth of rainfall of that area. There are many a type of gauging instruments available. Some of them are non-recording type for example., Symon's Rain gauge, Indian Standard Rain gauge (I.S Gauge) etc., and there are recording type of rain gauges for example., Tipping Bucket Rain Gauge, Weighing Bucket Type, Float Type etc., There are certain difficulties which come in the way of installing the gauging instruments. In urban areas it is difficult to locate sites free from the effect of winds and clear from obstructions. In remote location it is next to impossible to install a well-equipped gauging instrument and to monitor it. Since there is necessary for estimating the flood in these areas for planning and development works if any calls for the necessary of developing unit hydrograph. This calls for a study on development of unit hydrograph by synthetic/artificial

methods. The unit hydrographs developed by these methods are termed as the synthetic/artificial unit hydrograph.

There are many experimental studies are taken in order to develop a synthetic unit hydrograph. Some of them are developed by Snyder (1938), SCS Curve Number method, Clarks IUH (1945), McCarthy, and many more.

1.2 Snyder's Method

F F Snyder established a methodology to develop unit hydrograph in 1938, a most commonly used method to develop synthetic unit hydrograph. He analysed a large number of hydrographs from basin in Appalachian Mountain regions in United States, ranging from 25sq.km to 25,000 sq.km. To develop the unit hydrograph, it is necessary to know the time base, time lag, peak flow and width of UH. These parameters are sufficient to develop the unit hydrograph. The coefficients C_t and C_p are found to vary depending on the topology, geology and climate.

1.3 Why Snyder's Method?

Snyder's method is based on the basin morphological details and the three major features of a standard unit hydrograph. The hydrograph features are duration of effective rainfall, t_r , the peak rate of discharge, q_p and the basin lag time, T . From these an empirical relation is established to calculate the peak rate of discharge per unit area, q_{pR} , the basin time (t_b), and the widths of the unit hydrograph, W (in hours) at W_{50} i.e. 50% and W_{75} i.e. 75% of the peak discharge. With these values of Q_p , t_p , T , W_{50} and W_{75} a smooth graph to give 1 cm of runoff can be sketched. And thus, unit hydrograph can be developed in an easy and convenient way with out much errors in the process.

1.4 Methodology:

The Snyder's Method is employed in the present study to develop unit hydrograph for ungauged catchment area. The physiographic parameters are obtained by delineating the watershed using *ArcMap 10.3*. The Digital Elevation Model was successfully used to create shapefiles necessary to calculate the physiographic parameters. The lengths were calculated using *Google Earth Pro* by overlapping the shape file and the area was calculated using attribute table of the shape file. The Snyder's parameters are calculated and the graph is sketched.

1.5 Objectives:

To understand the type of hydrometeorological data required to develop Unit Hydrograph for an ungauged catchment.

1. To study morphological analysis using ArcGIS software.
2. To calculate Snyder's parameters.
3. To develop Synthetic Unit Hydrograph by Snyder's Method.

CHAPTER 2

REVIEW OF LITERATURE

2.1 Sudhakar BS, Anupam K S & Akshay OJ, 2015, “Snyder UH and GIS for Estimation of Flood for ungauged Catchments in Lower Tapi basin, India”

“The SUH method has been chosen for estimation of the flood response, sub-watershed parameters such as river length, length of centroid, spatial area, lateral slope and terrain and soil factors for estimating discharge. The geo database developed in ArcGIS environment from toposheet, a DEM of 50m cell size has been produced and sub-watershed characterisation was completed. The SUH model has been validated for peak discharge at one of the gauged sites.”

2.2 Abhay S Wayal, Dr. P V Parameswaran & Dr. N K Ameta, 2008, “Derivation of Unit Hydrograph for Ungauged Catchments”

This gives a summary of the Snyder’s relation for synthetic unit hydrograph for ungauged basins. “The UH parameters such as basin lag, peak discharge, base time, widths at 50% & 75% peak flow, were calculated from the developed relationship to draw UH. The synthetic derived based on Snyder’s method is employed.”

2.3 Azhar Husain, 2017, “Development of Synthetic Unit Hydrograph at Kakkadavu Dam”

This paper describes the application of hydrometeorological approach method for developing a SUH at the dam site it also gives a one-day probable maximum flood hydrograph. For the development of SUH where RF data is not available, empirical equations are employed depending on the basin characteristics.

2.4 P K Bhunya, S N Panda & M K Goel, 2011, “Synthetic Unit Hydrograph Methods: A critical review”

“Many different methods are reviewed in this study. The traditional methods e.g., Snyder’s Method, SCS Method and TS Method are checked along with Probability Distribution Functions based SUH methods. The study inferred that the Snyder’s and SCS methods were widely applied in practical engineering problem.”

2.5 Subhash Chander & Hari Shanker, 1984, “Unit Hydrograph based forecast model”

The paper presents a procedure for the online determination of phi-index and rainfall excess as a storm progress in time. This enables the application of a UH in real time river flow forecasting.

2.6 Nanjundi, Inaythulla, Chalapathi & Abhishek, 2016, “Hydrologic Analysis of Suryanagar Watersheds using IDF Curves and Runoff models”

This study said that due to nonlinear interactions between rainfall and runoff, synthetic relations are required for the estimation of the runoff hydrographs. The Suryanagar watershed considered here is studied for basin characteristics using Arc GIS.

2.7 Olalekan Adekunle, “An assessment of Digital Elevation Models from different spatial data sources”

DEM representing a very important geospatial data type in the analysis and modelling of different hydrological models. Different sources of spatial data in generation of DEMs are SRTM, Digitised Topographical map & Google Earth Pro. The study compared all these types. DEMs from toposheet though gave good result but the human error is notable. SRTM & Google Earth Pro. derived DEMs were reliable as there is absence of field work.

CHAPTER 3

MATERIALS AND METHODOLOGY

3.1 Study Area:

The watershed selected for the present study is located between Latitude $12^{\circ} 59' 59''$ N - $13^{\circ} 02' 23''$ N and Longitude $77^{\circ} 30' 59''$ E - $77^{\circ} 33' 57''$ E. The study area covers an area of 11.21 sq.km. The watershed consists of Yeshwanthpura suburb, Laggere, Nandini Layout and Mahalakshmi Puram, Bengaluru, Karnataka, India. The summer temperatures range from 21° C to 34° C and the winter temperatures ranges from 15° C to 30° C. The average annual total rainfall is about 905mm with around 108 rainy days per year. Thus, Bengaluru relishes a wholesome weather all around the year.

The area of the catchment is found out by delineating the Digital Elevation Model (DEM) downloaded from *USGS Earth Explorer* using ArcGIS software package. The area of the catchment is found to be 11.21 sq.km.

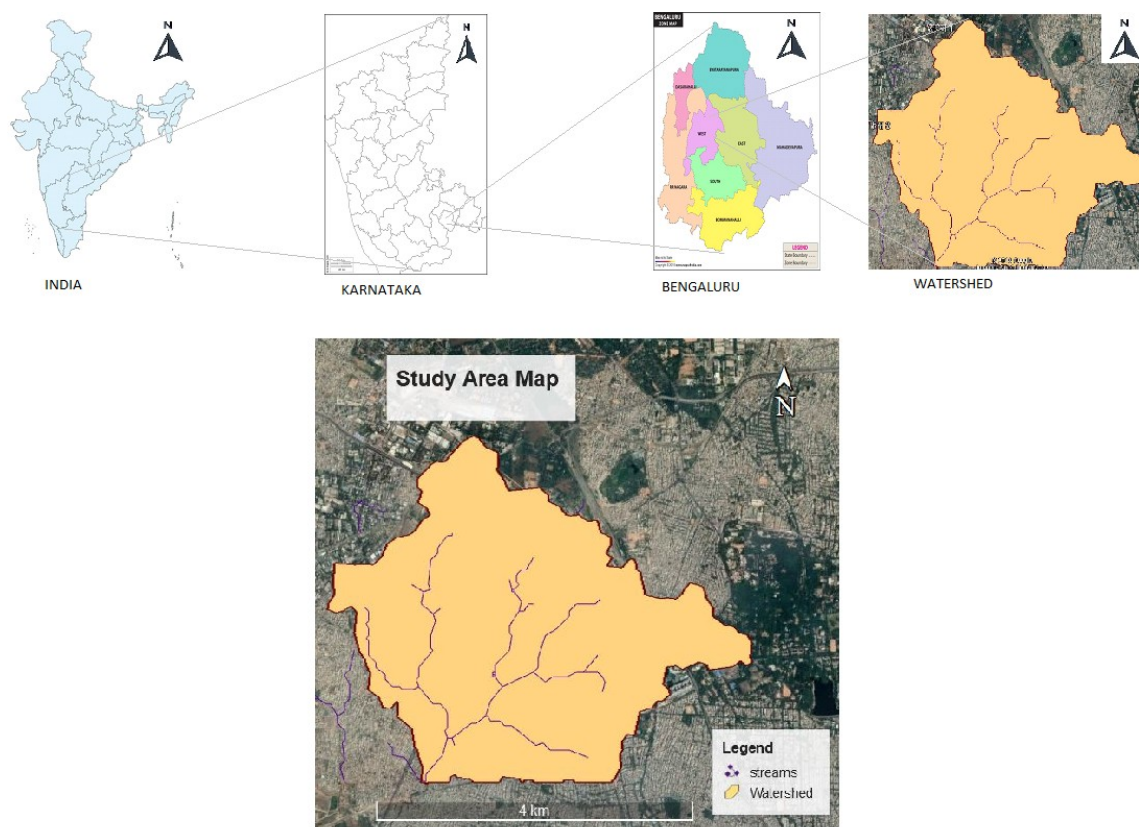


Figure 3.1 Location Map of Study Area

3.2 ArcGIS:

ArcGIS is a cloud-based mapping and analysis package. It is a Geographic Information System for working with maps and geographic information. A component of ArcGIS called as ArcMap 10.3 is used to create maps, manage geographic data, perform analysis. ArcMap has many a set of analysis tools, which are employed to perform certain functions. Every analysis tool creates a layer file in the table of contents, this layer files signifies geographic data in ArcMap software such as a particular theme of data. Example map layers include streams and lakes, political boundaries, roads, terrain, building footprints, utility lines. A base map/base layer is used for locational position and provides an outline on which users overlay or mashup their operational layers, perform tasks, and envision geographic data.

3.3 Digital Elevation Model (DEM)

A Digital Elevation Model is well-defined as "any digital representation of the continuous variation of relief over space," (Burrough 1986), where relief denotes the elevation of ground with respect to the datum considered. It can also be considered as frequently spread out grids of the elevation data, used for the unremitting spatial illustration of any terrain.

A Digital Elevation Model (DEM) is a digital geographic data in raster form. The land elevations of ground positions are sampled at regularly spaced horizontal intervals. DEM is generated from the elevation information from several points, which may be regular or irregular over the space. In the initial days DEMs were used to be developed from the contours mapped in the topographic maps and stereoscopic aerial images. Now we can directly download DEMs from the websites such as *USGS Earth Explorer* and several other websites.

3.4 Downloading Digital Elevation Model:

There are many sources to download a DEM, one such is Earth Explorer by United States Geological Survey. The Earth Explorer (EE) user interface is an online search tool developed by the United States Geological Survey (USGS).

4.4.1 Steps to download DEM:

1. Open USGS Earth Explorer website (<https://earthexplorer.usgs.gov>).
2. Register and login with an account.
3. In the “Search Criteria” search for the area of interest or use map points to draw polygon over the area of interest.
4. Choose data ranges (usually summer months are preferred).

Search Criteria | Data Sets | Additional Criteria | Results

1. Enter Search Criteria

To narrow your search area: type in an address or place name, enter coordinates or click the map to define your search area (for advanced map tools, view the [help documentation](#)), and/or choose a date range.

Address/Place | Path/Row | Feature | Circle

nandini layout bengaluru

Show Clear

Click on an Address/Place to show the location on the map and add coordinates to the Area of Interest Control.

Num	Address/Place	Latitude	Longitude
1	Nandini Layout, Bengaluru, Karnataka, India	13.0137	77.5339

Coordinates | Predefined Area | Shapefile | KML

Degree/Minute/Second | Decimal

i No coordinates selected.

Use Map Add Coordinate Clear Coordinates

Date Range | Result Options

Search from: mm/dd/yyyy to: mm/dd/yyyy

Search months: (all)

Data Sets » Additional Criteria » Results »

Figure 3.4.1 Entering Search Criteria

5. In “Data Sets” go to “Digital Elevation” and select “SRTM” under this select “SRTM 1Arc-Second Global” option.

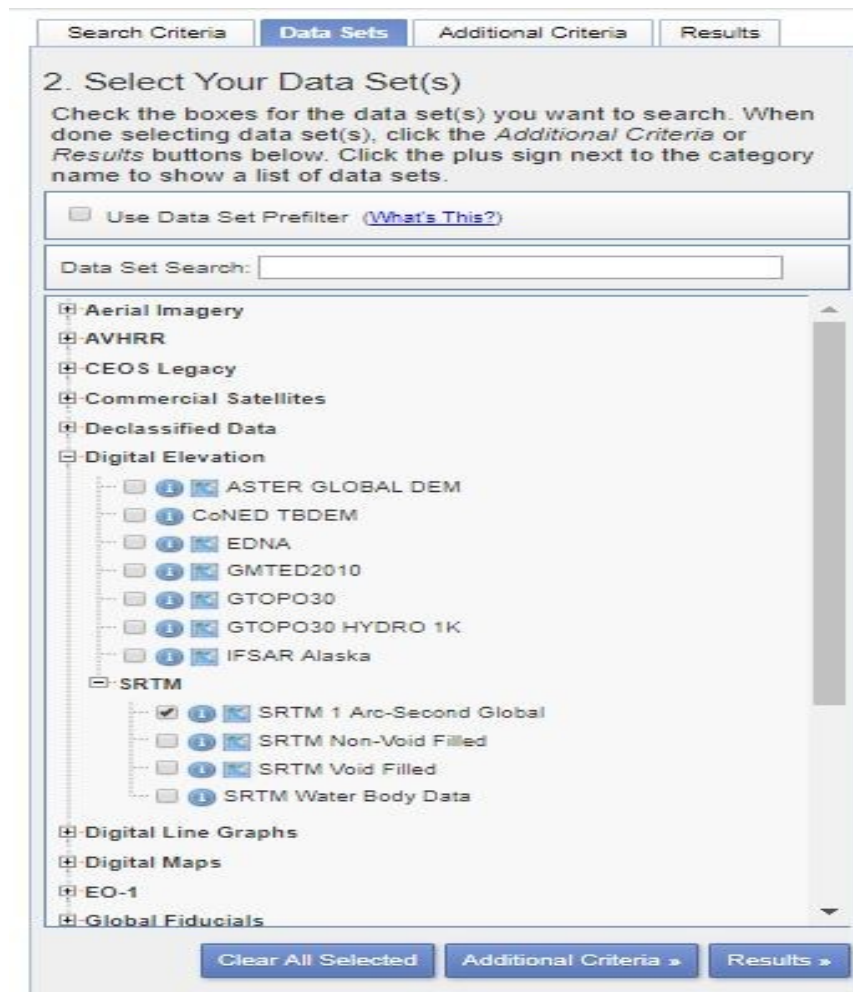


Figure 3.4.2 Selecting Data Sets

6. Click results, available DEMs will be show.



Figure 3.4.3 DEMs shown

7. Use show footprint or show overlay to select suitable DEM.

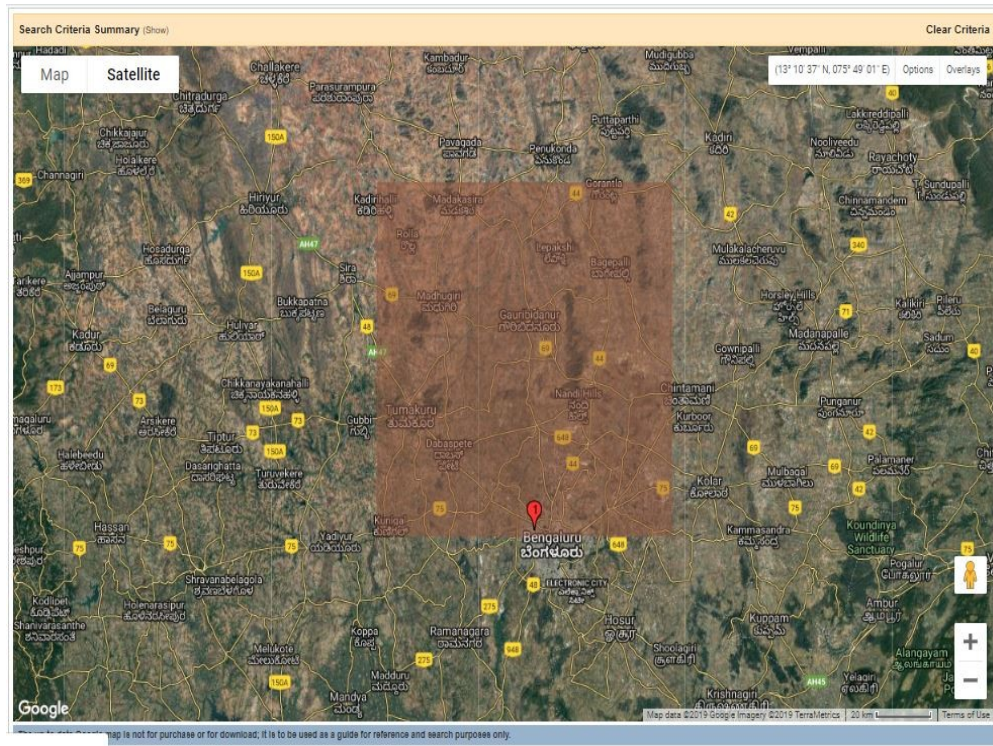


Figure 3.4.4 Foot Print of DEM

8. Download the suitable DEM out of available options.



Figure 3.4.5 DEM Download Options in Various Formats

The selected DEM for the study is published on 23 September 2014, with 1-Arc resolution, entitled with ID No. *SRTM1N13E077V3*

3.5 Delineation

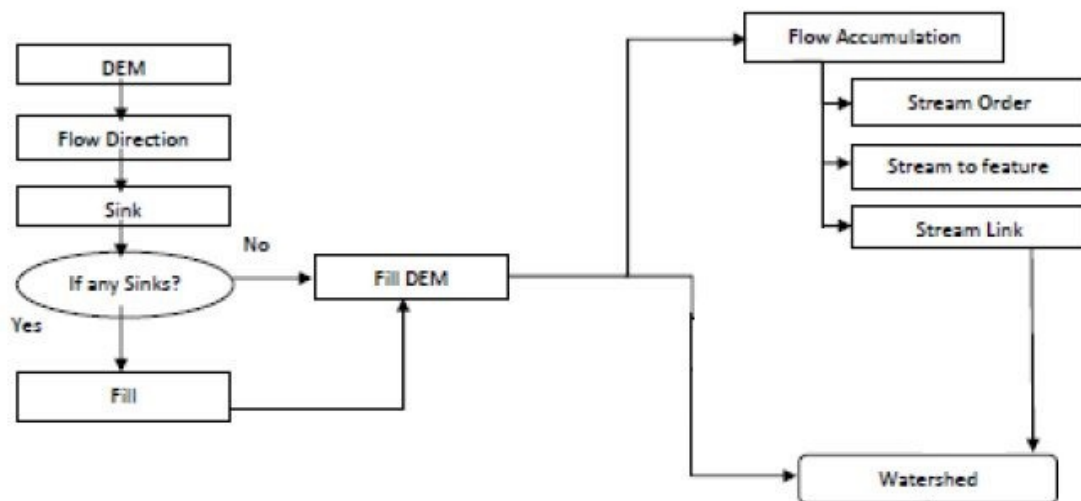


Figure 3.5.1 Flowchart of Delineation

Watershed are physically delineated by the area upstream from a specified outlet point or the mouth. Watersheds can be delineated manually using toposheets or digitally in a GIS environment. The procedure said below comprises step by step explanation for delineating basins in ESRI's ArcGIS Desktop 10.3 software package.

After downloading the required DEM from the USGS Earth Explorer website for the area of interest. Start with the delineation process.

Step 1: Setup the work directory

1. Open the ArcMap and create a new blank map document.
2. Use the *Add Data* option to incorporate the base layer for geo referencing.
3. Use the *Add Data* option to incorporate the DEM for the area of interest.
4. Save the map document (*File>Save*).

Step 2: Opening *Hydrology* toolbox

Open *Arc Toolbox* from the main menu and expand *Spatial analysis Tools > Hydrology* toolbox to view its contents.

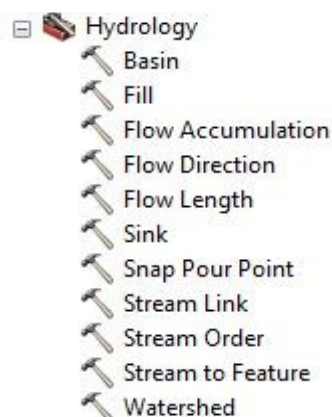


Figure 3.5.2 Hydrology Toolbox

Step 3: Creating the DEM for the necessary area

Since the DEM is spread for a large area over different area, we need to select for required area only. Use *Rectangle* option in *Drawing* menu for selecting the area roughly. Then in the map view select *Data* option and go to *Export Data* (Right click on Dem Layer > Data>Export Data > Save) this will be saved as a new layer. We can now delete the previous DEM layer (right click on the map layer > remove). Save the map document.

Step 4: Create a depression less DEM

The *Fill* in the Hydrology toolbox is used to remove the imperfections or sinks if any in the DEM. A sink/imperfection is a cell in DEM that does not have an associated discharge value.

1. Double click on the *Fill* tool select the DEM as the *Input surface raster*
2. The output file can be given a specified name and can be saved in the specified file directory.

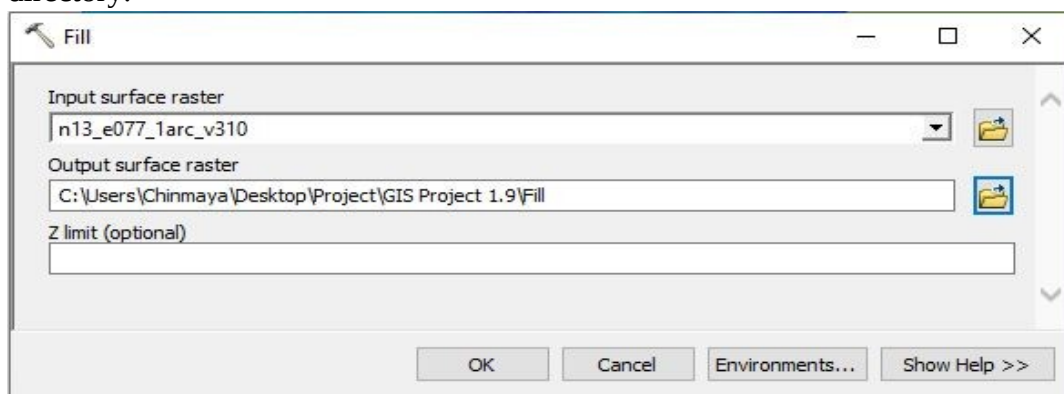


Figure 3.5.3 Fill Tool

3. Click *OK* to run the tool. It may take some time depending on the software and processor speeds. A dialogue in the right bottom will appear after successful completion of the tool.

Step 5: Create a Flow Direction

1. Double click on the *Flow Direction* tool, select the layer from previous step as the *Input surface raster* (i.e., Fill)
2. The output file can be given a specified name and can be saved in the specified file directory.

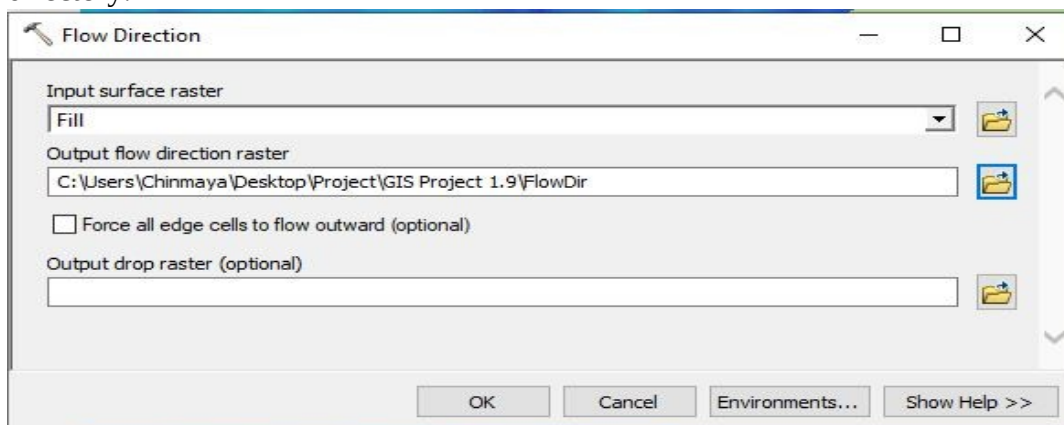


Figure 3.5.4 Flow Direction Tool

- Click *OK* option to run the tool. A dialogue in the right bottom will appear after successful completion of the tool.

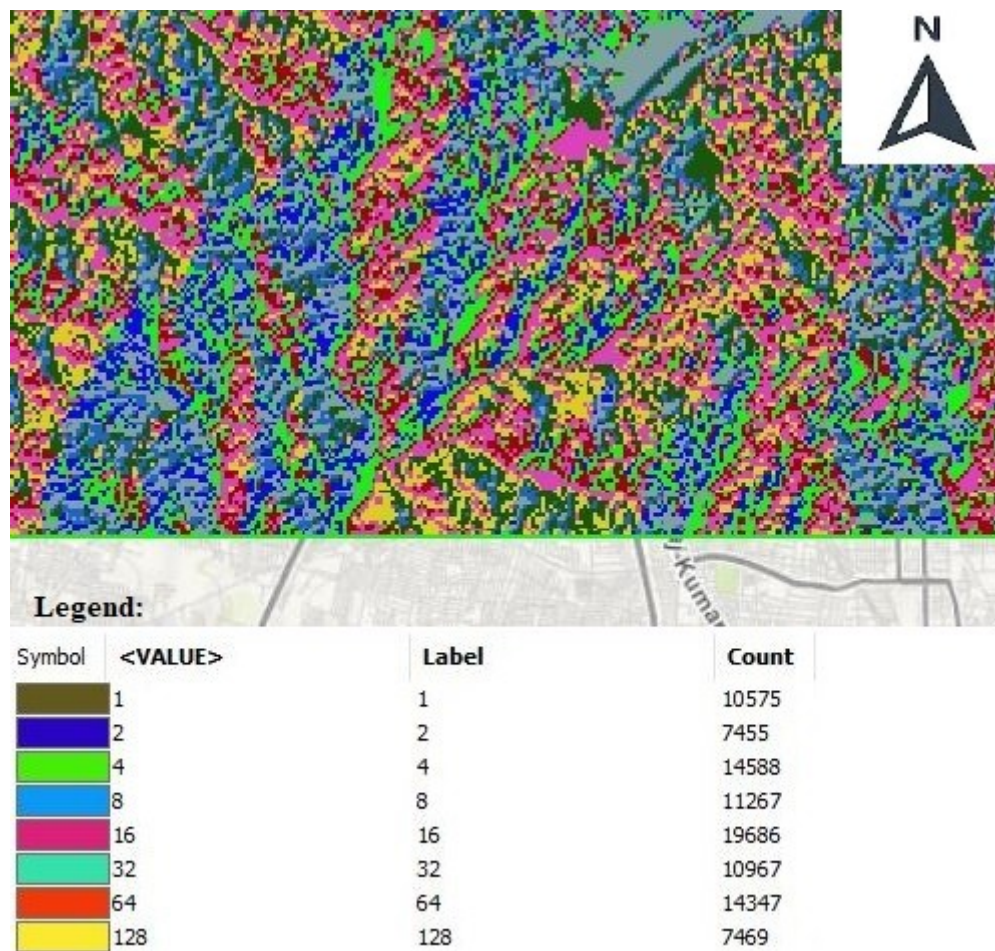


Figure 3.5.5 Output of Flow Direction Tool

Step 5: Create Flow Accumulation Grid

- Double click on the *Flow Accumulation* tool, select the layer from previous step as the *Input surface raster* (i.e., FlowDir)
- The output file can be given a specified name and can be saved in the specified file directory.
- The output data type to be selected as integer type.

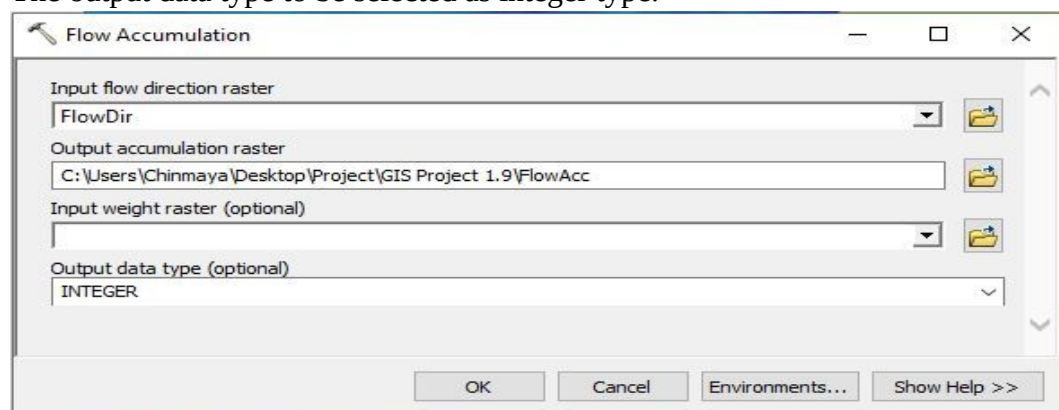


Figure 3.5.6 Flow Accumulation Tool

4. Click **OK** to run the tool. A dialogue in the right bottom will appear after successful completion of the tool.

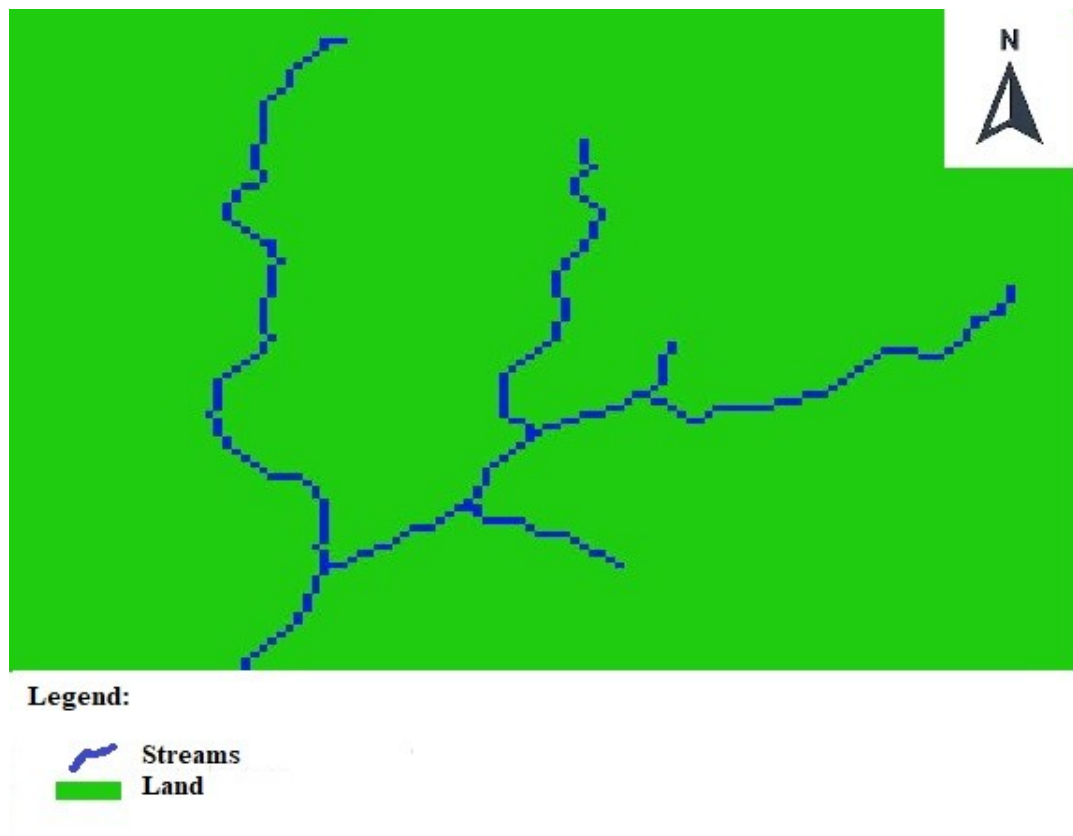


Figure 3.5.7 Output of Flow Accumulation Tool

5. Double click on the new layer formed (*FlowAcc*) in the map view.
6. In symbology, for classification select *Classes* as 2 and minimum value of break value as 1000. Click **OK**.

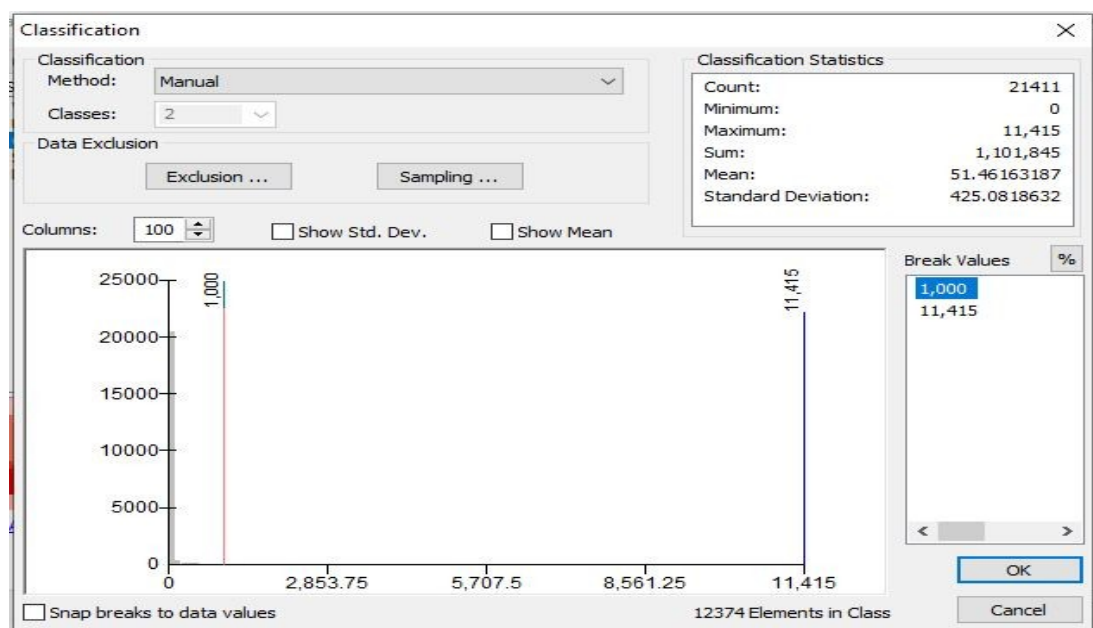


Figure 3.5.8 Classification of Flow Accumulation

Step 6: Basin tool

This tool creates a raster delineating all the drainage basins.

1. Double click on the *Basin* tool, select the FlowDir layer from *Flow Direction* tool output as the *Input surface raster*.
2. The output file can be given a specified name and can be saved in the specified file directory.

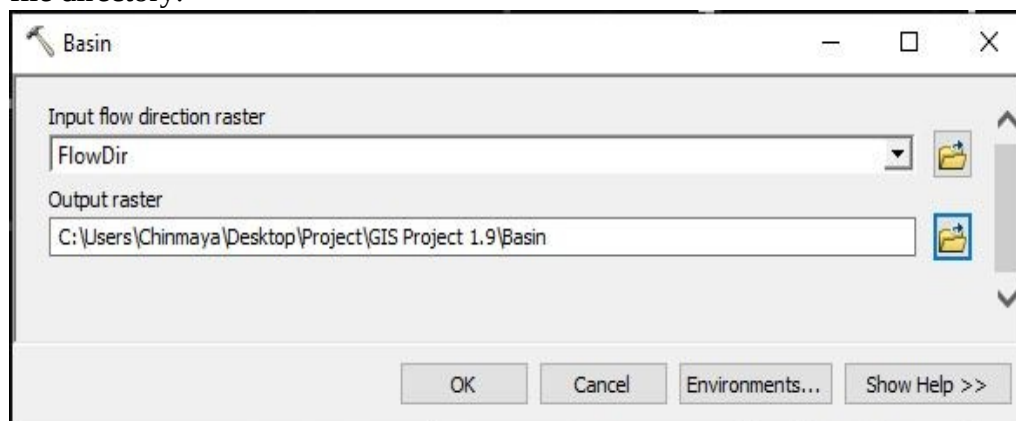


Figure 3.5.9 Basin Tool

3. Click *OK* to run the tool. A dialogue in the right bottom will appear after successful completion of the tool.

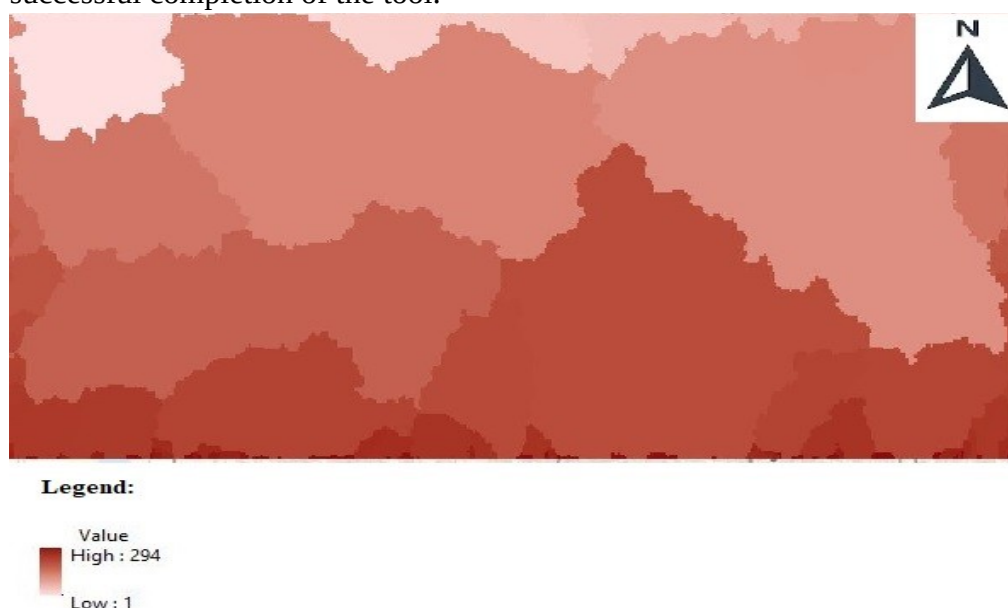


Figure 3.5.10 Output of Basin Tool

Step 7: Conversion tool (Raster to Polygon)

This tool is used to converted the raster datasets to polygon data. The basin the raster form will be converted to polygon shapes.

1. In Conversion Tools select “*From Raster to Polygon*” tool and double click on it.
(*Conversion Tools > From Raster > From Raster to Polygon*)
2. Select the Basin layer from *Basin* tool output in the last step as the *Input surface raster*.

3. The output file can be given a specified name and can be saved in the specified file directory.

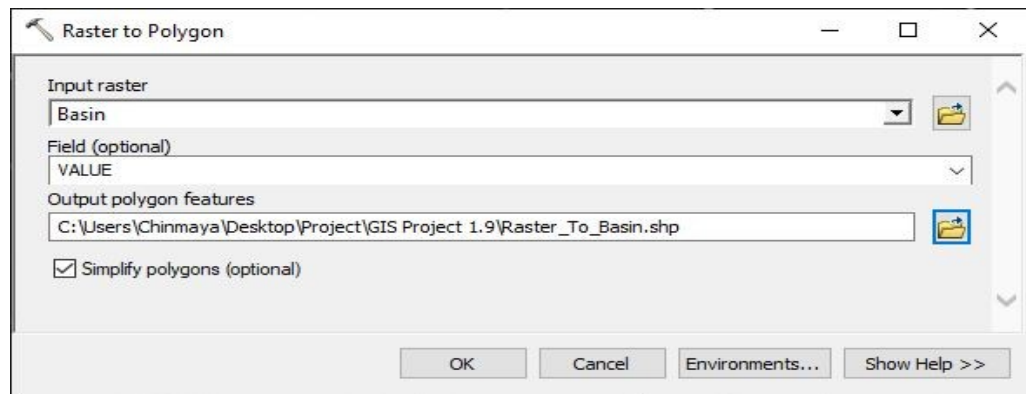
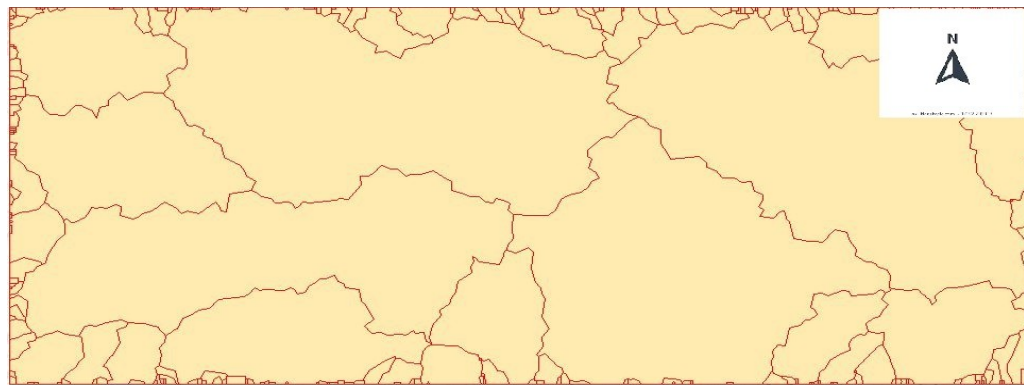


Figure 3.5.11 Raster to Polygon Tool

4. Click OK to run the tool. A dialogue in the right bottom will appear after successful completion of the tool.



Legend:

- Sub cathcmnts**
- Borders**

Figure 3.5.12 Output of Raster to Polygon Tool

Step 8: Clip tool

This tool is used to clip the selected sub-catchment out of many in the available in watershed.

1. In Geo Processing menu select Clip tool.
2. The output shape file from the previous step is taken as *Input Feature* and *Clip Feature*.

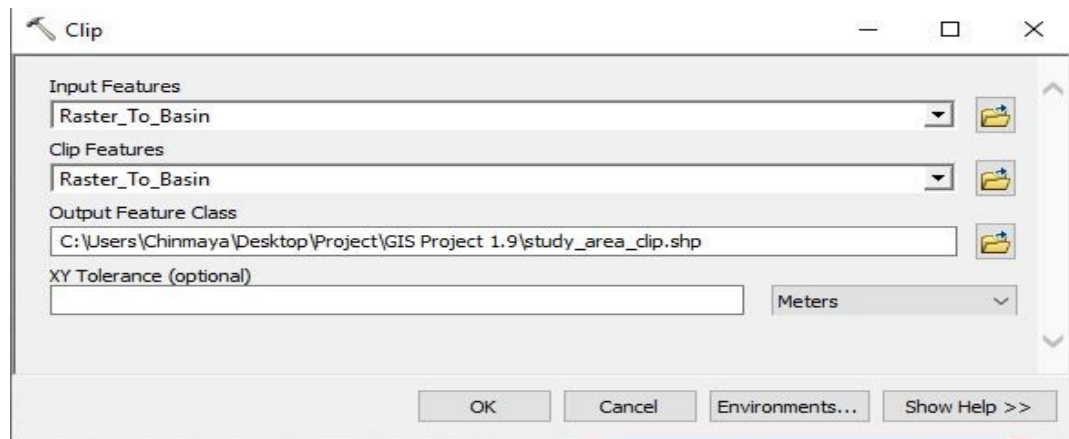


Figure 3.5.13 Clip Tool

3. Accept all other defaults options and click **OK** to run the tool.



Figure 3.5.14 Clipped Watershed

Step 9: Raster Calculator

This tool permits us to create and perform a Map Algebra calculation that will output a raster. This is employed to narrow down the flow value to values greater than 1000 for the study area (basin selected)

1. Go to *ArcTool Box > Map Algebra > Raster Calculator*
2. Define algebraic equation. In *Layers and Variables* select *FlowAcc* shape file and define the algebraic expression of flow value greater than 1000

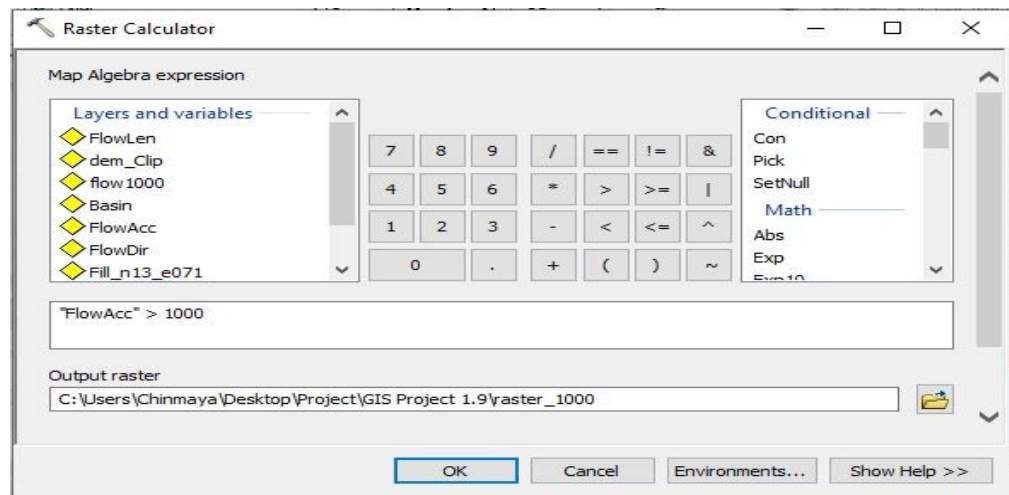


Figure 3.5.15 Raster Calculator Tool

3. Click *OK* to run the tool. A dialogue in the right bottom will appear after successful completion of the tool a dialogue box will appear in the right bottom corner.

Step 10: Raster to Polyline Conversion tool

This tool is used to convert the flow streams in the raster data to polyline data form.

1. In Conversion Tools select *From Raster to Polygon* tool and double click on the tool.
(*Conversion Tools > From Raster > From Raster to Polyline*)
2. Select the shapefile from the previous step as the *Input surface raster*.
3. The output file can be given a specified name and can be saved in the specified file directory.

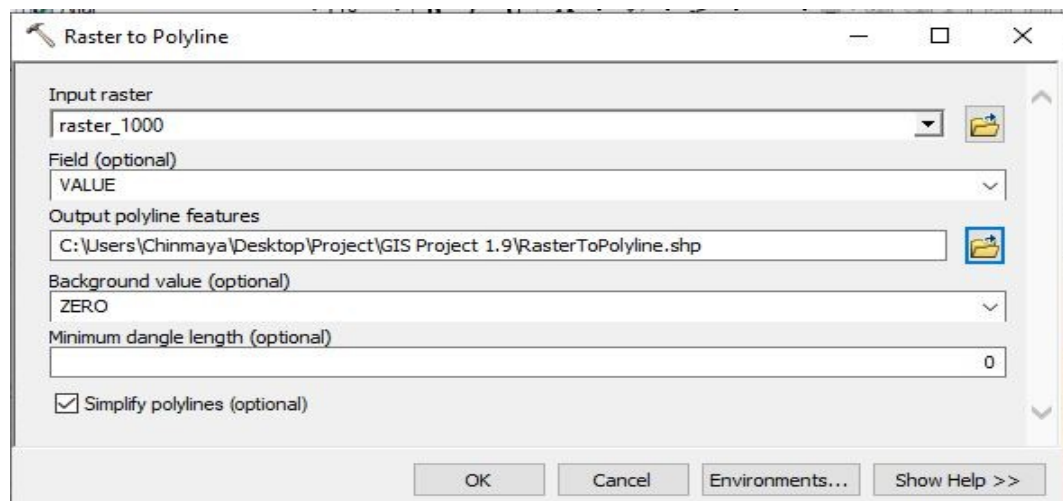


Figure 3.5.16 Raster to Polyline Tool

- Accept all other defaults and click *OK* to run the tool. A dialogue in the right bottom will appear after successful completion of the tool.

Step 11: Clipping the study area.

To clip the study area against the flow raster generated in the last step.

- Go *Arc Tool > Data Management Tools > Raster > Clip*.
- Select the DEM as the *Input Raster* and the output shape file from Step 10 is taken as *Output Extent*.

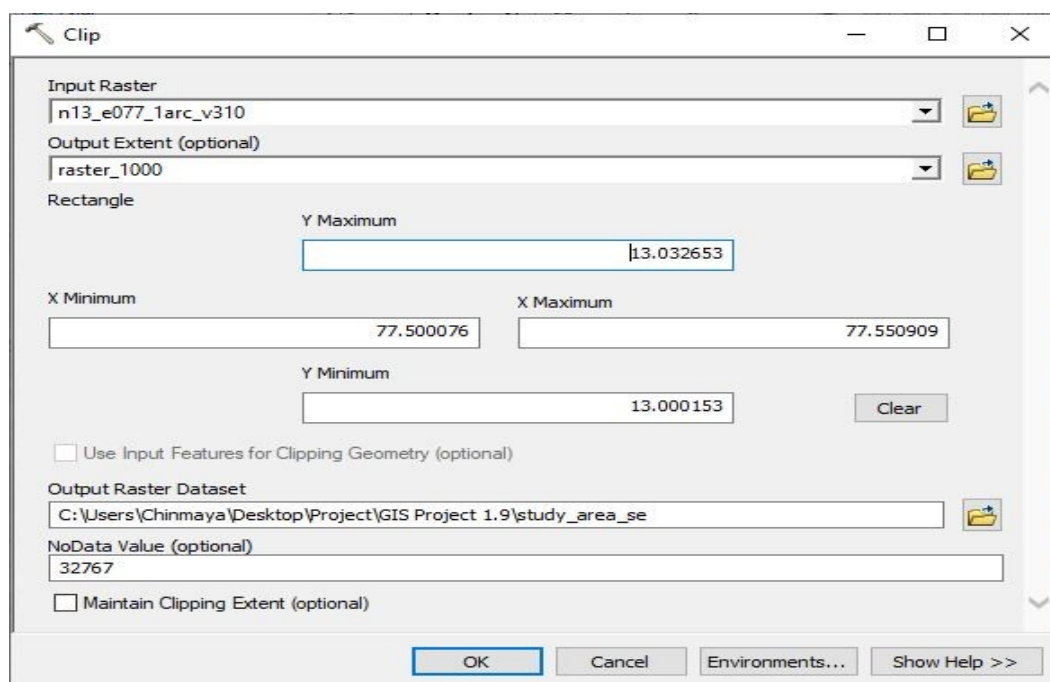


Figure 3.5.17 Clipping the Watershed

- Accept all other defaults and click *OK* to run the tool. A dialogue in the right bottom will appear after successful completion of the tool.
- This tool separates the study area from the whole DEM.

3.6 Calculation of Flow Parameters:

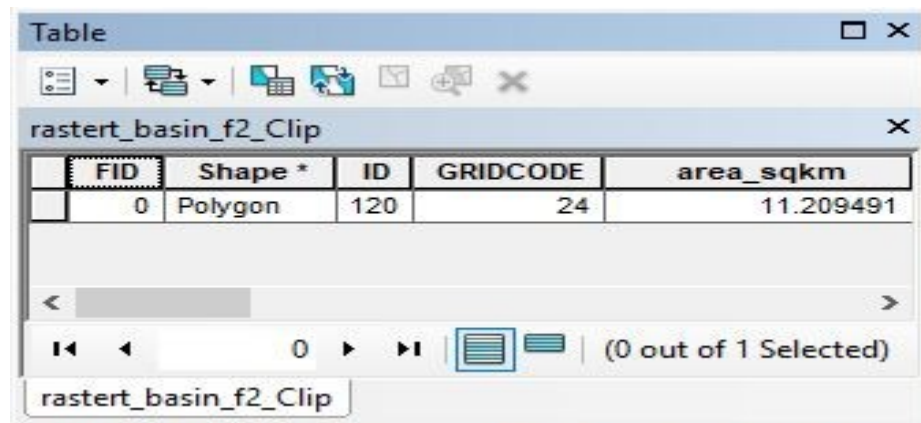
3.6.1 Area of the Catchment:

The area of the catchment is calculated using the *Calculate Geometry* in the *Attribute Table* of the shape file created in the Step 11.

Steps to Calculate Area using Attribute Table in ArcMap.

- Open the Attribute Table by right clicking on the shape file output in the Step.11

2. In table options add field, and name it as Area.
3. Right click on the Area field created in the previous step.
4. Select Calculate Geometry.
5. Select the property as Area and desired units are selected.
6. Click OK. The tool will result in the area calculation and the area is displayed in the Area column in the *Table*.



FID	Shape *	ID	GRIDCODE	area_sqkm
0	Polygon	120	24	11.209491

Figure 3.6.1 Attribute Table

3.7 Calculation of stream length and centroid length:

To calculate the length of the stream and the centroid length we use Google Earth Pro. The shape files in the ArcMap are converted to KMZ files and opened with Google Earth Pro and it can be observed that the KMZ files directly overlaps to the exact place of the catchment.

3.7.1 Steps to Convert Shape file to KMZ file:

1. Go to *Arc Tool Box* menu open *Conversion Tool* open *To KML* select *Layer To KML*.
2. Select the basin shapefile created in Step 11 as the *Input surface raster*.
3. The output file can be given a specified name and can be saved in the specified file directory.

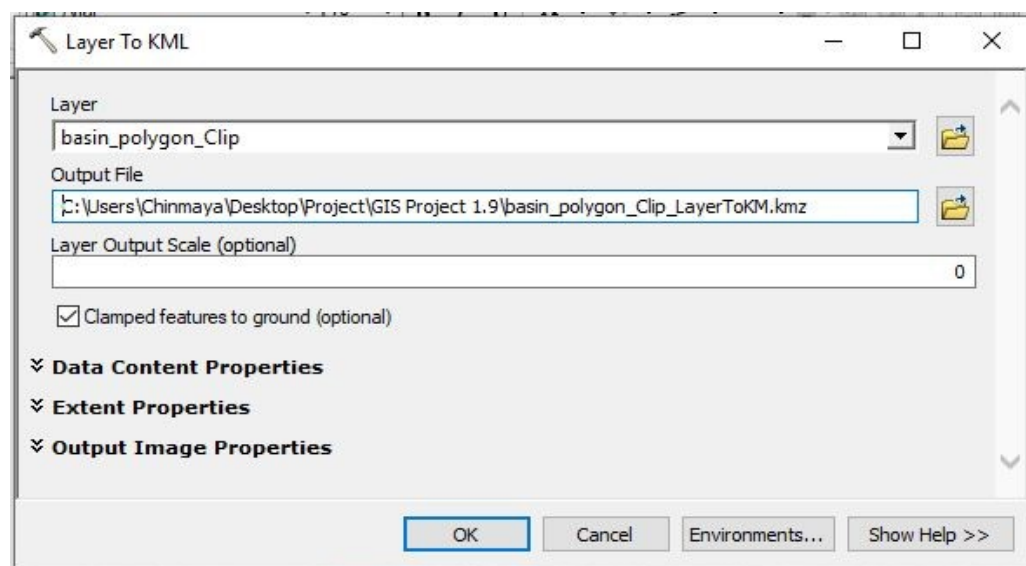


Figure 3.7.1 Layer to KML Tool

4. Accept all other defaults and click *OK* to run the tool. A dialogue in the right bottom will appear after successful completion of the tool.
5. Open the KMZ file. This will redirect to *Google Earth Pro*

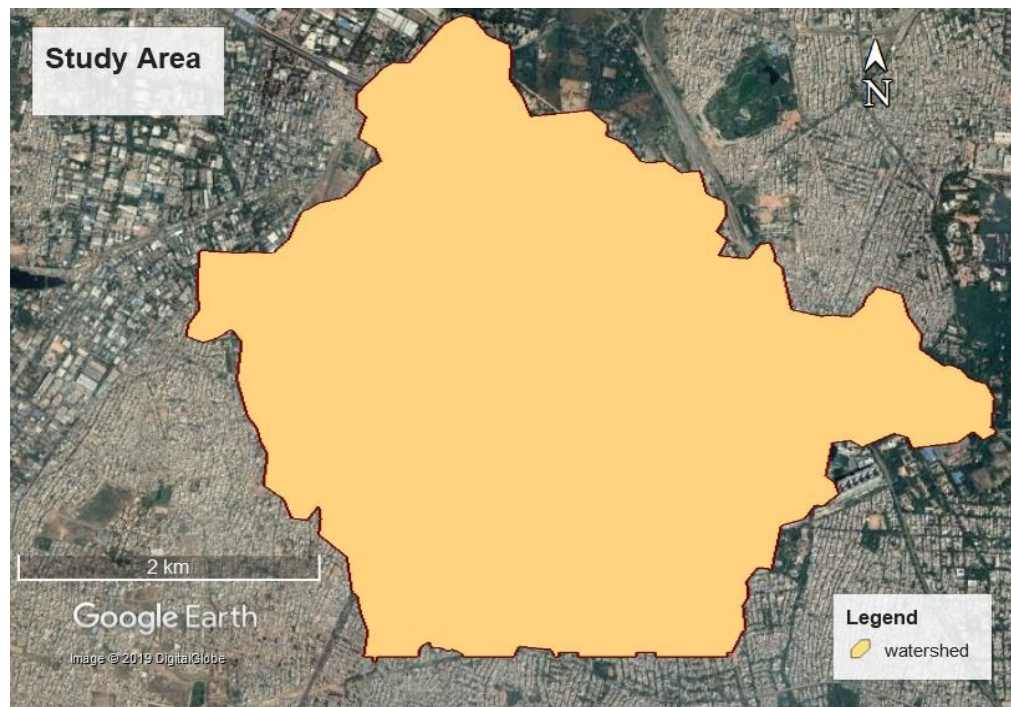


Figure 3.7.2 KMZ File of Watershed in *Google Earth Pro*

The same procedure is followed to convert the Flow Streams shape files to KMZ files.

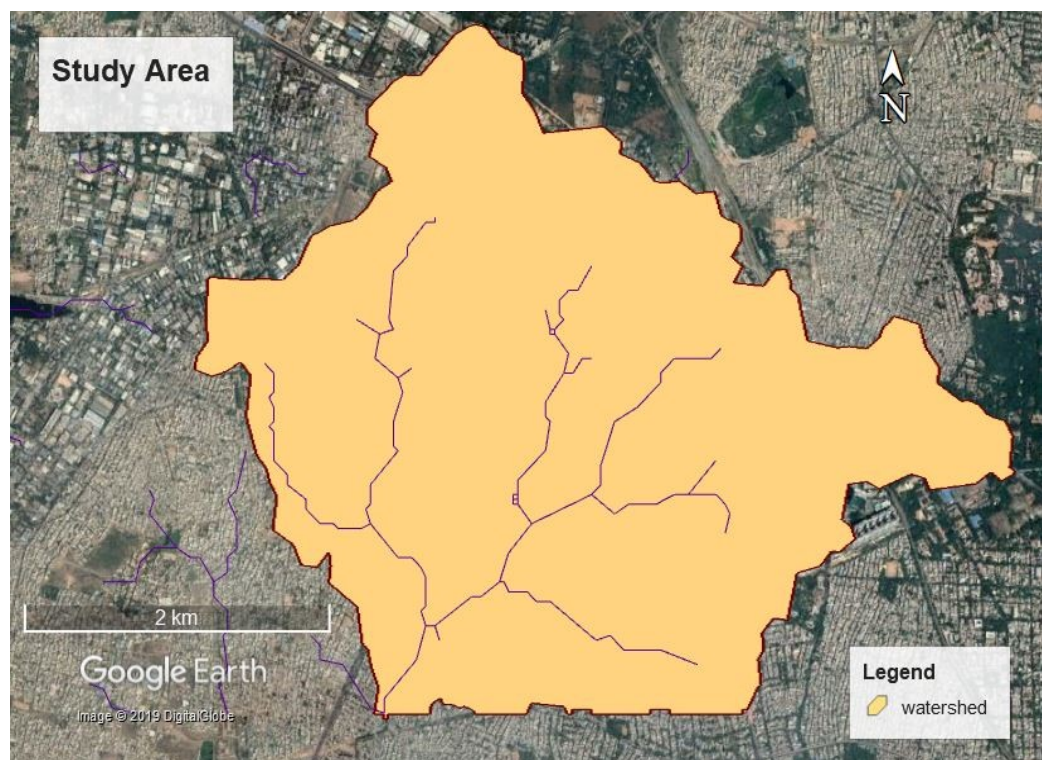


Figure 3.7.3 KMZ File of Watershed with streams in *Google Earth Pro*

3.7.2 Steps to Calculate length of the Streams:

1. The *Ruler* tool is used to measure the length of the stream.
2. Go to *Path* option.
3. The length is measured along with the streams with multiple points on the ground.
4. Start from the mouth of the watershed and go on measuring till the end of the stream.
5. The same procedure is followed to measure all the stream lengths. However, the longest stream length is considered for the further calculation of Snyder's parameters.

3.7.3 Calculating the centroid Length L_C :

To calculate the centroid length of the stream we use Google Earth Pro. However, to find the centroid point we incorporated *MASSPROP* tool in AutoCAD.

3.7.4 Steps to Convert Shape file to DWG file:

1. Go to *Arc Tool Box* menu open *Conversion Tool* open *To CAD* select *Export to CAD*.
2. Select the basin shapefile created in Step 11 as the *Input surface raster*.
3. The output file can be given a specified name and can be saved in the specified file directory.
4. Accept all other defaults and click *OK* to run the tool. A dialogue in the right bottom will appear after successful completion of the tool.
5. Open the CWG file. This will redirect to *AutoCAD*.

The shape file in the ArcMap is converted into DWG file and opened with AutoCAD. The *MASSPROP* command is used and the coordinates of the centroid is given in the respective units. We then retraced the centroid using *POINT* command by giving the coordinates obtained. This file is then converted into KMZ file again using conversion tools. Then the file is opened in Google Earth Pro with centroid being marked already.

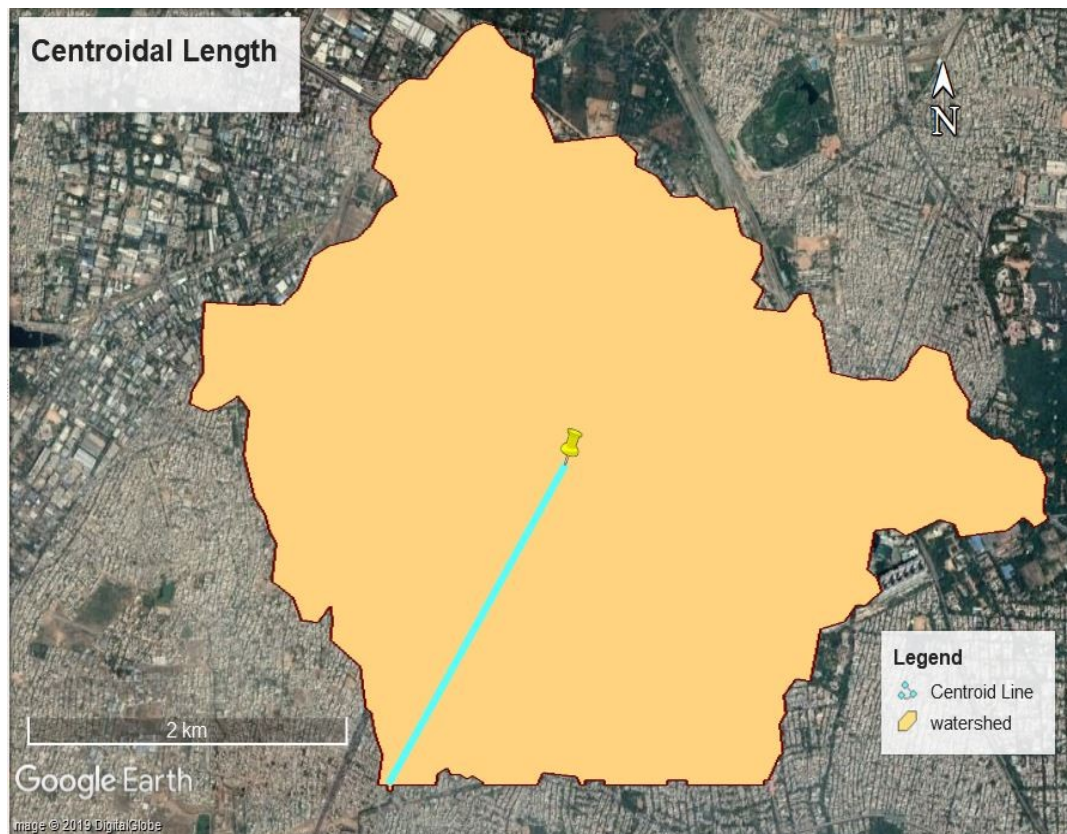


Figure 3.7.4.1 Centroidal Length

The centroid length is calculated by following the same procedure as to calculate the length of streams.

Thus, all the flow parameters are calculated.

The flow parameters are:

Parameter	Value
Area	11.209491 ~ 11.209 sq.km
Length (L)	3.82 km
Centroid Length (L_C)	2.13 km

Table 3.1 Flow Parameters

CHAPTER 4

Results and Analysis:

4.1 Calculation of Snyder's Parameters:

The Snyder's parameters are calculated using the Physiographic parameters such as area of the catchment, the length of the stream and the centroidal length along with the regional constants C_p and C_t obtained. However, the regional constants are obtained from the same hydro meteorologically homogeneous area from the know basin parameters.

1. The basin lag: (t_p)

The time interval from the centroid of rainfall axis to the runoff produced in the catchment is the basin lag. It is given by the equation:

$$t_p = C_t (LL_C)^{0.30}$$

2. Peak Discharge: (Q_p)

It is the peak discharge seen in the Unit Hydrograph or the maximum rate of discharge during the period of runoff caused by a storm. It is given by:

$$q_p = (2.778 \times C_p) / t_p$$

3. The base width: (T)

The base width of the Unit Hydrograph in hours. It is given by

$$T = 72 + 3t_p$$

4. The standard duration of the Unit Hydrograph was fixed by Snyder as

5. Alternatively, the base time of the Synthetic Unit Hydrograph can be found using the fact that the area under the Unit Hydrograph is equivalent to a direct runoff of 1 cm. assuming a triangular shape for the Unit Hydrograph, the base time from this principle is given by

$$T' = 5.556 / q_p$$

6. If a synthetic Unit Hydrograph for any other duration D' is required, then the modified basin lag time is given by

$$t_p' = t_p + (D' - D) / 4$$

7. The modified lag time t_p' is then used in the equation to calculate the standard

$$q_p$$

$$\text{i.e. } q_p' = (2.778 \times C_p) / t_p'$$

$$\text{i.e. } Q_p = (2.778 \times A \times C_p) / t_p'$$

8. Similarly, the base width/base lag of the D' hour unit Hydrograph is obtained from the previous equation by replacing t_p by t_p' .
9. The sketching of the Synthetic Unit Hydrograph is made easier by calculating the width of the Hydrograph at a rate of discharge equal to certain per cent of peak discharge. They are given by:

$$W_{50} = 2.14(q_p')^{-1.08}$$

$$W_{75} = 1.22(q_p')^{-1.08}$$

Where W_{50} and W_{75} are the widths of the hydrograph in hours at 50% and 75% of the peak discharge respectively. Usually it is considered that $1/3^{\text{rd}}$ of this width is distributed beforehand the peak of unit hydrograph and the $2/3^{\text{rd}}$ after the peak.

4.2 Calculations:

Considering the duration of Unit Hydrograph as 4 hours i.e. $D' = 4\text{h}$, and assuming that the peak discharge occurs at 24h

4.2.1 The basin lag:

$$t_p' = t_p + (D' - D)/4$$

$$t_p' = t_p + 3/4$$

$$\text{where, } t_p' = 24 - (D'/2)$$

$$t_p' = 24 - (4/2) = 22 \text{ h}$$

$$t_p = 24.0952 \text{ h}$$

4.2.2 The regional constant C_t :

The regional constant C_t is calculated using the formula

$$C_t = t_p / (LLC)^{0.3}$$

$$C_t = 24.0952 / (3.82 \times 2.13)^{0.3} = 12.8468$$

4.2.3 The Peak Discharge:

$$Q_p' = (2.778 \times A \times C_p) / t_p'$$

$$Q_p' = (2.778 \times 11.209 \times 0.57) / 24.0952$$

$$Q_p' = 0.7366 \text{ cumecs.}$$

4.2.4 The Discharge per unit Area:

The Discharge per unit area is given by

$$q_p' = Q_p' / A$$

$$q_p' = 0.0657 \text{ cumecs/sq.km}$$

4.2.5 The modified Base time/Base width T':

$$T' = 5.556 / q_p'$$

$$T' = 5.556 / 0.0657$$

$$T' = 84.5662 \text{ h}$$

4.2.6 W_{50} and W_{75} :

The widths of the Unit Hydrograph at 50% and 75% of the peak discharge are:

$$\text{a. } W_{50} = 2.14(q_p')^{-1.08}$$

$$W_{50} = 2.14(0.0657)^{-1.08}$$

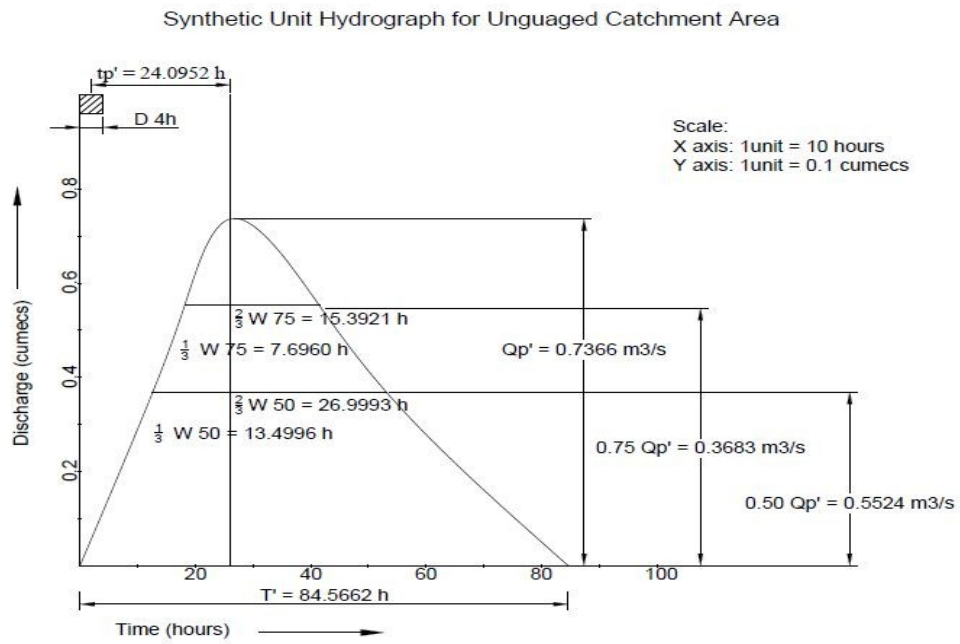
$$W_{50} = 40.4989 \text{ h.}$$

$$\text{b. } W_{75} = 1.22(q_p')^{-1.08}$$

$$W_{75} = 1.22(0.0657)^{-1.08}$$

$$W_{75} = 23.0882 \text{ h.}$$

With these values of Q_p' , t_p' , T' , W_{50} and W_{75} a smooth graph to give one cm of runoff can be sketched.



Graph 4.2.1 4h Synthetic Unit Hydrograph of Study Area

5. Conclusion:

1. The aim of this study was to develop unit hydrograph for any ungauged catchment.
2. The project was based on Snyder's Method of Synthetic Hydrograph.
3. The study was mainly concentrated on delineating the watershed and calculating the physiographic parameters further calculation of Snyder's parameters and developing the unit hydrograph.

Thus, unit hydrograph of ungauged catchment was developed.